

DATA STRUCTURES - IT23302 HW ASSIGNMENT SET - 1

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Batch : IT Batch II Second Year

1. Sum of 100 matrices.

Code

```
2. #include<stdio.h>
3. int main()
4. {
5.     int matrix1[100][100],row,column,matrix2[100][100],sum[100][100];
6.     printf("Enter the number of Rows : ");
7.     scanf("%d",&row);
8.     printf("Enter the number of columns : ");
9.     scanf("%d",&column);
10.    printf("Array 1 elements\n");
11.    for(int i=0;i<row;i++)
12.    {
13.        for(int j=0;j<column;j++)
14.        {
15.            scanf("%d",&matrix1[i][j]);
16.        }
17.    }
18.    printf("Array 2 element\n");
19.
20.    for(int i=0;i<row;i++)
21.    {
22.        for(int j=0;j<column;j++)
23.        {
24.            scanf("%d",&matrix2[i][j]);
25.        }
26.    }
27.    for(int i=0;i<row;i++)
28.    {
29.        for(int j=0;j<column;j++)
30.        {
31.            sum[i][j]=matrix1[i][j]+matrix2[i][j];
32.        }
33.    }
34.    printf("Resulting Matrix from addition\n");
```

```
35. for(int i=0;i<row;i++)
36. {
37.     for(int j=0;j<column;j++)
38.     {
39.         printf("%d ",sum[i][j]);
40.     }
41.     printf("\n");
42. }
43. return 0;
44.
45. }
```

Output

```
• PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 1.c -o 1
• PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 1.c -o 1.exe
• PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./1.exe
Enter the number of Rows : 2
Enter the number of columns : 2
Array 1 elements
1
1
1
1
Array 2 element
2 2
2 2
Resulting Matrix from addition
3 3
3 3
❖ PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> |
```

2. Write a program in C to read n numbers of values in an array and display them in reverse order

```
#include <stdio.h>
int main()
{
    int arr[100],n;
    printf("Enter the Number of elements in the array : ");
    scanf("%d",&n);
    printf("Array Elements");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    printf("Printing in reverse Order : \n");
    for(int i=n-1;i>=0;i--)
    {
        printf("%d ",arr[i]);
    }
    return 0;
}
```

OUTPUT

```
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 2.c -o 2
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 2.c -o 2.exe
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./2.exe
Enter the Number of elements in the array : 3
Array Elements
3
4
Printing in reverse Order :
4 3 2
```

3. Count the total number of duplicate elements in an array.

CODE

```
//Count the total number of duplicate elements in an array
#include<stdio.h>
int main()
{
    int arr[100],n,count=0;
    printf("Enter the Number of elements in the array : ");
    scanf("%d",&n);
    printf("Array Elements");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n-1;i++)
    {
        if(arr[i]==arr[i+1]) count++;
    }
    printf("The total number of duplicates in the array : %d",count);
    return 0;
}
```

OUTPUT

```
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./3.exe
Enter the Number of elements in the array : 4
Array Elements1
1
3
2
The total number of duplicates in the array : 1
```

4. Print unique elements in an array

CODE

```
#include <stdio.h>
int main()
{

    int arr[100], n, count = 0,temp;
    printf("Enter the Number of elements in the array : ");
    scanf("%d", &n);
    printf(" Array Elements");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n-i-1;j++)
        {
            if(arr[j]>arr[j+1])
            {
                temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }

    printf("Unique Elements in the array : \n");
    for (int i = 0; i < n; i++)
    {
        if (arr[i] == arr[i + 1] && i!=n-1)
            continue;
        printf("%d ", arr[i]);
    }
    return 0;
}
```

OUTPUT

```
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 4.c -o 4
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 4.c -o 4.exe
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./4.exe
Enter the Number of elements in the array : 4
Array Elements2
3
5
6
Unique Elements in the array :
2 3 5 6
```

5. Insert in sorted array.

CODE

```
#include <stdio.h>
int main()
{
    int arr[100], n, low=0, high, target, mid, ind;
    printf("Enter the Number of elements in the Sorted array: ");
    scanf("%d", &n);
    high=n-1;
    printf("Enter Array Elements in sorted manner");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Element to be inserted : ");
    scanf("%d", &target);
    while(low<=high)
    {
        mid=low+(high-low)/2;
        if(arr[mid]>target)
        {
            ind=mid;
            high=mid-1;
        }
        else low=mid+1;
    }
    for(int i=n;i>ind;i--)
    {
        arr[i]=arr[i-1];
```

```

    }
    arr[ind]=target;
    printf("After insertion \n");
    for (int i = 0; i <=n; i++)
    {
        printf("%d ", arr[i]);
    }
return 0;
}

```

OUTPUT

```

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 5.c -o 5.exe
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./5.exe
Enter the Number of elements in the Sorted array: 4
Enter Array Elements in sorted manner2
3
5
6
Element to be inserted : 4
After insertion
2 3 4 5 6
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> 

```

6. Insert in unsorted array.

CODE

```

#include <stdio.h>
int main()
{

    int arr[100], n,target,ind;
    printf("Enter the Number of elements in the array: ");
    scanf("%d", &n);
    printf("Array Elements");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Element to be inserted : ");

```

```

scanf("%d",&target);
printf("Element's index : ");
scanf("%d",&ind);

for(int i=n;i>ind;i--)
{
    arr[i]=arr[i-1];
}
arr[ind]=target;
printf("After insertion \n");
for (int i = 0; i <=n; i++)
{
    printf("%d ", arr[i]);
}
return 0;
}

```

OUTPUT

```

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 6.c -o 6.exe
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./6.exe
Enter the Number of elements in the array: 5
Array Elements2
3
5
6
8
Element to be inserted : 7
Element's index : 3
After insertion
2 3 5 7 6 8

```

7. Matrix multiplication.

CODE

```

#include<stdio.h>
int main()
{
    int matrix1[100][100],row1,column1,row2,column2,sum,matrix2[100][100],multi[100][100];
    printf("Enter the number of Rows in matrix1 : ");
    scanf("%d",&row1);
    printf("Enter the number of columns in matrix1 : ");
    scanf("%d",&column1);

```

```
printf("Array 1 elements\n");
for(int i=0;i<row1;i++)
{
    for(int j=0;j<column1;j++)
    {
        scanf("%d",&matrix1[i][j]);
    }
}
printf("Enter the number of Rows in matrix2 : ");
scanf("%d",&row2);
printf("Enter the number of columns in matrix2 : ");
scanf("%d",&column2);
printf("Array 2 element\n");

for(int i=0;i<row2;i++)
{
    for(int j=0;j<column2;j++)
    {
        scanf("%d",&matrix2[i][j]);
    }
}
for(int i=0;i<row1;i++)
{
    for(int j=0;j<column2;j++)
    {
        sum=0;
        for(int k=0;k<row2;k++)
        {
            sum+=matrix1[i][k]*matrix2[k][j];
        }
        multi[i][j]=sum;
    }
}
printf("Resulting Matrix from Multiplication\n");
for(int i=0;i<row1;i++)
{
    for(int j=0;j<column2;j++)
    {
        printf("%d ",multi[i][j]);
    }
}
```

```

    }
    printf("\n");
}
return 0;
}

```

OUTPUT

```

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 7.c -o 7.exe
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./7.exe
Enter the number of Rows in matrix1 : 2
Enter the number of columns in matrix1 : 3
Array 1 elements
1 2 3
4 4 4
Enter the number of Rows in matrix2 : 3
Enter the number of columns in matrix2 : 3
Array 2 element
1 1 1
1 1 1
1 1 1
Resulting Matrix from Multiplication
6 6 6
12 12 12

```

8. Matrix power.

CODE

```

#include<stdio.h>
int main()
{
    int matrix1[100][100],row1,column1,transpose[100][100];
    printf("Enter the number of Rows in matrix1 : ");
    scanf("%d",&row1);
    printf("Enter the number of columns in matrix1 : ");
    scanf("%d",&column1);

    printf("Array 1 elements\n");
    for(int i=0;i<row1;i++)
    {
        for(int j=0;j<column1;j++)
        {
            scanf("%d",&matrix1[i][j]);
        }
    }
}

```

```

}
//Transposing
for(int i=0;i<row1;i++)
{
    for(int j=0;j<column1;j++)
    {
        transpose[j][i]=matrix1[i][j];
    }
}
printf("Resulting Matrix from transpose\n");
for(int i=0;i<column1;i++)
{
    for(int j=0;j<row1;j++)
    {
        printf("%d ",transpose[i][j]);
    }
    printf("\n");
}
return 0;
}

```

OUTPUT

```

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 8.c -o 8
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> gcc 8.c -o 8.exe
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./8.exe
Enter the number of Rows in matrix1 : 3
Enter the number of columns in matrix1 : 3
Array 1 elements
1 1 1
2 2 2
3 3 3
Resulting Matrix from transpose
1 2 3
1 2 3
1 2 3
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1>

```

9. Right diagonal sum.

CODE

```

#include<stdio.h>
int main()

```

```

{
    int matrix1[100][100],row1,column1,sum=0;
    printf("Enter the number of Rows in matrix1 : ");
    scanf("%d",&row1);
    printf("Enter the number of columns in matrix1 : ");
    scanf("%d",&column1);

    printf("Array 1 elements\n");
    for(int i=0;i<row1;i++)
    {
        for(int j=0;j<column1;j++)
        {
            scanf("%d",&matrix1[i][j]);
        }
    }
    for(int i=0;i<row1;i++)
    {
        sum+=(matrix1[i][row1-i-1]);
    }
    printf("The Right Diagonal Sum : %d",sum);
    return 0;
}

```

OUTPUT

```

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./9.exe
Enter the number of Rows in matrix1 : 3
Enter the number of columns in matrix1 : 3
Array 1 elements
1 0 0
0 1 0
0 0 1
The Right Diagonal Sum : 1

```

10. Check for identity matrix.

CODE

```

#include<stdio.h>
int main()
{
    int matrix1[100][100],row1,column1,transpose[100][100],sum=0,found;
    printf("Enter the number of Rows in matrix1 : ");
    scanf("%d",&row1);

```

```

printf("Enter the number of columns in matrix1 : ");
scanf("%d",&column1);

printf("Array 1 elements\n");
for(int i=0;i<row1;i++)
{
    for(int j=0;j<column1;j++)
    {
        scanf("%d",&matrix1[i][j]);
    }
}
for(int i=0;i<row1;i++)
{
    if(matrix1[i][i]!=1)
    {
        printf("The Given Matrix is not a identity matrix");
        found=-1;
        break;
    }
}
if(found!=-1) printf("The Given Matrix is a identity matrix");
return 0;
}

```

OUTPUT

```

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./10.exe
Enter the number of Rows in matrix1 : 3
Enter the number of columns in matrix1 : 3
Array 1 elements
1 0 0
0 1 0
0 0 1
The Given Matrix is a identity matrix
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./10.exe
Enter the number of Rows in matrix1 : 2
Enter the number of columns in matrix1 : 2
Array 1 elements
1 2 3 3
The Given Matrix is not a identity matrix

```

11. Find the number of vowels, consonants, digits and white spaces.

CODE

```

#include<stdio.h>
int main()
{
    char str[100],temp;
    int vowels=0,cons=0,digits=0,ws=0;
    printf("Enter the string : ");
    fgets(str,sizeof(str),stdin);
    for(int i=0;str[i]!='\0';i++)
    {
        temp=str[i];
        if(temp == 32)
        {
            ws++;
        }
        else if(temp>=48 && temp<=57) digits++;
        else if(temp=='a' ||temp=='e' ||temp=='i' ||temp=='o' ||temp=='u' ||temp=='A' ||temp=='E'
||temp=='I' ||temp=='O' ||temp=='U')
        {
            vowels++;
        }
        else if((temp>='a'&&temp<='z') || (temp>='A'&&temp<='Z')) cons++;
    }

    printf("Vowels = %d\n",vowels);
    printf("Consonants = %d\n",cons);
    printf("Digits = %d\n",digits);
    printf("White Spaces = %d\n",ws);
}

```

OUTPUT

```

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./11.exe
Enter the string : Rahul
Vowels = 2
Consonants = 3
Digits = 0
White Spaces = 0

```

12. Find the frequency of a character in a string.

CODE

```

#include<stdio.h>
int main()

```

```

{
    char str[100],ch;
    int count=0;
    printf("Enter the String : ");
    fgets(str,sizeof(str),stdin);
    printf("Character to be counted : ");
    scanf("%c",&ch);
    for(int i=0;str[i]!='\0';i++)
    {
        if(str[i]==ch) count++;
    }
    printf("The Frequency of the given Character is %d ",count);
    return 0;
}

```

OUTPUT

```

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./12.exe
Enter the String : V Rahul
Character to be counted : a
The Frequency of the given Character is 1

```

13. Insert words in a string.

CODE

```

#include<stdio.h>
int main()
{
    char str[100];
    int count=0;
    printf("Enter the String : ");
    fgets(str,sizeof(str),stdin);
    for(int i=0;str[i]!='\0';i++)
    {
        if (str[i]!=32 && str[i]!='\n' && (i==0 || str[i-1]==32)) count++;
    }
    printf("The number of words in the given String : %d",count);
    return 0;
}

```

OUTPUT

```
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./13.exe
Enter the String : Rahul Krishna
The number of words in the given String : 2
```

14. Sort string array.

CODE

```
#include<stdio.h>
#include<string.h>
int main()
{
    char names[100][100],temp[100];
    int row;
    printf("Enter the Number of names : ");
    scanf("%d",&row);
    for(int i=0;i<row;i++)
    {
        scanf("%s",names[i]);
    }
    for(int i=0;i<row;i++)
    {
        for(int j=0;j<row-i-1;j++)
        {
            if(strcmp(names[j],names[j+1])>0)
            {
                strcpy(temp,names[j]);
                strcpy(names[j],names[j+1]);
                strcpy(names[j+1],temp);
            }
        }
    }
    printf("Sorted names : ");
    for(int i=0;i<row;i++)
    {
        printf("%s ",names[i]);
    }
}
```

OUTPUT

```
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./14.exe
Enter the Number of names : 3
Rahul
Kishore
Abhi
Sorted names : Abhi Kishore Rahul
```

15. Toggle case of a sentence (lowercase to uppercase and vice versa).

CODE

```
#include<stdio.h>
int main()
{
    char str[100];
    int count=0;
    printf("Enter the String : ");
    fgets(str,sizeof(str),stdin);
    for(int i=0;str[i]!='\0';i++)
    {
        if(str[i]>='A' && str[i]<='Z')
        {
            str[i]=str[i]+32;
        }
        else if(str[i]>='a' && str[i]<='z')
        {
            str[i]=str[i]-32;
        }
    }
    printf("The toggled String = %s",str);
    return 0;
}
```

OUTPUT

```
PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_1> ./15.exe
Enter the String : Rahul
The toggled String = rAHUL
```